PARTNERSHIP MEMO

Japan–Korea Strategic Energy & Industrial Partnership

Where deeper industrial integration creates strategic value, through which instruments, and where to stay out

Generated from Allied Industrial Base Explorer data and a curated external evidence layer (accessed 2026-11-06); evidence, sources and caveats in the appendix. Data as of 30 October 2026.

Executive Decision Summary

BUILD TOGETHER

Solar Manufacturing: Wafers (Japan->Korea). Permanent Magnets & Drive Systems: Rare-earth oxides & compounds (joint); Rare-earth permanent magnets (Japan->Korea). Battery Materials & Cells: Natural graphite (joint); Cobalt intermediates & chemicals (joint); Lithium chemicals (joint); Battery materials (foil, electrolyte) (Japan->Korea). Grid Equipment & Conductors: Electric Grid refining (joint). Semiconductors & Power Electronics: Electric Grid components (Japan->Korea).

TRADE AND STANDARDS

Recognise trusted partner supply of electric vehicles refining (rare-earth metals, rare-earth oxides & compounds) in procurement and content rules. Common anti-circumvention rules for batteries refining (battery materials (foil, electrolyte), cathode materials & precursors, cobalt intermediates & chemicals) and wind refining (copper wire & rod, rare-earth metals, rare-earth oxides & compounds) and rare-earth oxides & compounds and cobalt intermediates & chemicals and lithium chemicals. Bind market access for wafers, rare-earth permanent magnets. Keep pgms (iridium, ruthenium); copper ores & concentrates; nickel intermediates & chemicals open to global sourcing.

INSTITUTIONS

Upgrade Korea-Japan Supply Chain Partnership Arrangement and Industrial and Trade Policy Dialogue: sector strategy, reciprocal commitments, a KPI dashboard and an annual review. No new MOU.

STAY OUT

Do not co-subsidise new Coal components capacity. Do not co-subsidise new Accumulator parts & separators capacity. Do not co-subsidise new Coal assembly capacity.

Why this partnership matters.

- Of 142 technology x stage cells and electro-industrial nodes assessed for the pair, 33 show an active complementarity (a supplier meeting the other's need, a shared chokepoint, a capacity-constrained trusted supplier or a chain specialisation), 6 direct competition and 78 no strategic case. The relationship is not one of general similarity: most of the electro-industrial base needs no bilateral intervention.[1]
- AIB strategic roles: Japan Capital & Investment Anchor (primary), Specialized Technology & Manufacturing Anchor (primary), Scale Manufacturing & Integration Platform (primary), Coordination / Bridge Hub (secondary); Korea Specialized Technology & Manufacturing Anchor (primary), Scale Manufacturing & Integration Platform (primary), Coordination / Bridge Hub (secondary), Capital & Investment Anchor (secondary).[1]

The central bilateral bargain.

- *Inference.* Korea supplies solar cells, cathode materials & precursors, gallium, germanium & indium, non-oriented electrical steel (ngo) where Japan is short; Japan supplies wafers, rare-earth permanent magnets, battery materials (foil, electrolyte), piezo-electric quartz where Korea is short; the two share exposure to the

same coercive supplier in rare-earth oxides & compounds, natural graphite, cobalt intermediates & chemicals, lithium chemicals, polysilicon, where neither can supply the other and the answer is joint demand, finance and third-country capacity.

- *Inference.* From the AIB roles, Japan brings capital, specialised technology and manufacturing, scale manufacturing; Korea brings specialised technology and manufacturing, scale manufacturing, coordination across allied chains. Public finance runs through JBIC, NEXI, JOGMEC on the Japan side and KEXIM, K-SURE, KOMIR, KDB on the Korea side. The trade is market access and demand certainty for trusted capacity.

The 5 highest-priority joint bets (tier 1 by the rules in Section 3, one per compact; ranked by urgency then importance):

PRIORITY	SECTOR	NODES IN THE BET (DIRECTION)	STRATEGIC PROBLEM	JAPAN CONTRIBUTION	KOREA CONTRIBUTION	MAIN INSTRUMENT (LEAD NODE)	TIMING
1	Solar Manufacturing	Wafers (Japan->Korea)	geopolitical choke point	Wafers: trusted capacity (28% of world supply)	Wafers: demand and long-term offtake	anchor offtake	12-24 months
2	Permanent Magnets & Drive Systems	Rare-earth oxides & compounds (joint); Rare-earth permanent magnets (Japan->Korea)	geopolitical choke point	Rare-earth oxides & compounds: its demand for pooled offtake, and finance for non-Chinese supply; Rare-earth permanent magnets: trusted capacity (9% of world supply)	Rare-earth oxides & compounds: its demand for pooled offtake, and finance for non-Chinese supply; Rare-earth permanent magnets: demand and long-term offtake	anchor offtake	12-24 months
3	Battery Materials & Cells	Natural graphite (joint); Cobalt intermediates & chemicals (joint); Lithium chemicals (joint); Battery materials (foil, electrolyte) (Japan->Korea)	geopolitical choke point	Natural graphite: its demand for pooled offtake, and finance for non-Chinese supply; Cobalt intermediates & chemicals: its demand for pooled offtake, and finance for non-Chinese supply	Natural graphite: its demand for pooled offtake, and finance for non-Chinese supply; Cobalt intermediates & chemicals: its demand for pooled offtake, and finance for non-Chinese supply	anchor offtake	12-24 months
4	Grid Equipment & Conductors	Electric Grid refining (joint)	geopolitical choke point	Electric Grid refining: its demand for pooled offtake, and finance for non-Chinese supply	Electric Grid refining: its demand for pooled offtake, and finance for non-Chinese supply	anchor offtake	12-24 months
5	Semiconductors & Power Electronics	Electric Grid components (Japan->Korea)	strategic manufacturing gap	Electric Grid components: trusted capacity (top-quarter position)	Electric Grid components: demand and long-term offtake	market access	12-24 months

Sources: 1.

- *Inference.* Outside the electro-industrial compacts, the rules also find fuel-trade lanes (coal refining); they are handled in the trade agenda, not as compacts.

Do not prioritise (Section 10):

- *Recommendation.* Do not co-subsidise new Coal components capacity: both are strong suppliers (Japan 9%, Korea 6%). Trusted supply is adequate: this is commercial competition.
- *Recommendation.* Do not co-subsidise new Accumulator parts & separators capacity: both are strong suppliers (Japan 11%, Korea 13%). More trusted capacity is needed, so agree who specialises in what rather than racing.
- *Recommendation.* Do not co-subsidise new Coal assembly capacity: both are strong suppliers (Japan 5%, Korea 4%). Trusted supply is adequate: this is commercial competition.

Existing mechanisms and the MEA question (Section 8):

- *Inference.* MEA assessment: **upgrade an existing framework (JPKR-M1)**. Gaps: no cross-sector KPI dashboard or review cycle; no bilateral mechanism covers SC-SOLAR, SC-SEMIS; JPKR-M1 has no published project pipeline.
- *Recommendation.* Do not sign a new energy MOU. Give JPKR-M1 a sector strategy, reciprocal commitments and a KPI dashboard covering the recommended compacts, reviewed every 12 months.

1. Current State of the Relationship

This section sets out what already exists before anything new is proposed. Facts are from the curated evidence layer (accessed 2026-11-06); items marked partial or unverified rest on secondary sources and should be confirmed before citation.

Alliance and plurilateral architecture

- Korea's memberships in OSI's economic-security agreements table: IPEF, MSP, U.S. FTA.[1]
- DFAT notes RCEP entered into force for Korea on 1 Feb 2022 and may be used by exporters as an alternative to KAFTA; the member list shown here is partial.[2]
- The 16 Mar 2023 Tokyo summit restored shuttle diplomacy, with Japan pledging to lift its three-material curbs and Korea to drop its WTO case.[3]
- Korean and Japanese officials agreed a close cooperation system on critical-mineral supply risks, support for company and agency cooperation on joint development in third countries (Latin America, Asia, Africa), and policy coordination in the Minerals Security Partnership and IEA; no projects were named.[4]
- The dialogue, set up Feb 2024 with a first meeting in Jun 2024, held its second meeting in Tokyo on 26 Mar 2025 and agreed to develop a Japan-ROK cooperation platform for hydrogen and ammonia supply chains; the METI page could not be opened.[5]
- Senior officials meeting in New York agreed to enhance dialogue on critical minerals and cooperation in Southeast Asia and Sub-Saharan Africa through the trilateral Early Warning System.[6]
- At the October 2025 Tokyo conference the two business federations agreed Korea's CPTPP participation was important given US protectionism and supply-chain disruption.[7]
- The second round discussed a joint response to falling copper treatment and refining charges, cooperation within the Minerals Security Partnership and ways to develop critical minerals jointly in third countries; no agreement or named project was reported.[8]
- On 13 Jan 2026 the leaders discussed supply-chain cooperation in depth and agreed to launch agency-level talks on economic security cooperation, with rare-earth stabilisation reported as a focus; no outputs were published.[9]
- The US-hosted ministerial in Feb 2026 was attended by Japan and Korea among others, and Korea was reported to chair the FORGE critical-minerals forum through June 2026.[10]
- Korea chaired the Minerals Security Partnership until June 2026; a February 2026 report says its 17 members carried into the successor FORGE; Korean public lenders joined the MSP Finance Network; no signed US-Korea

critical-minerals MOU was found.[11]

- A standing consultative body launched 14 Mar 2026 to review trade cooperation, economic security, supply chains, steel and mineral resources; the same article reports plans for joint exploration and investment in critical minerals and shared market monitoring; no schedule published.[12]
- The 19 May 2026 trade-ministry release covers crude oil swaps and mutual supply in a crisis, deeper LNG cooperation under the KOGAS-JERA agreement, SCPA crisis mechanisms and regional reserves via Japan's proposed POWERR Asia; no projects or funding named; hydrogen and critical minerals not covered.[13]
- In June 2026 Korea's trade ministry said no decision had been made on applying to CPTPP after reports of a late-June application; Japan reportedly would back accession without first requiring Korea to lift its Fukushima seafood ban; the Japan position is a press report.[14]
- The joint statement backs coordination on resource procurement and supply-chain stability, support for Korea's CPTPP accession, hydrogen certification and safety standards, critical-mineral stockpiling and crisis response through the SCPA, and an early resumption of the Korea-US-Japan Business Dialogue; business statement, not binding.[15]

Trade architecture and frictions

- In the OSI energy basket, Japan imported US$10.2bn from Korea and Korea imported US$10.7bn from Japan (bilateral flows 2024, summed over the technology x stage cells).[1]
- RCEP entered into force for Japan on 1 Jan 2022 and for Korea on 1 Feb 2022, giving the two countries a shared preferential trade framework with no separate bilateral FTA; tariff schedules not reviewed.[16]
- From July 2019 Japan required individual licences for fluorinated polyimide, photoresist and hydrogen fluoride exports to Korea and removed Korea from its white list; Korea removed Japan from its own list in Aug 2019 and filed a WTO complaint; Japan lifted the three-item curbs in Mar 2023.[17]
- Korea completed procedures to drop its 2019 WTO complaint on 23 Mar 2023, a week after the 16 Mar Yoon-Kishida summit; Japan had pledged to lift the three-material curbs; the article does not confirm the curbs had by then been lifted.[18]

Industrial and energy agreements

- The three leaders agreed to pilot an early warning system exchanging supply-chain information via embassies in selected countries, with critical minerals and rechargeable batteries as priority products, plus an annual commerce and industry ministers track; the system is not published as operating beyond the pilot.[19]
- KNOC reportedly signed an MOU with JOGMEC in February 2025 on cooperation in hydrogen, ammonia and CCS.[20]
- Ministers Kim Jung-kwan and Akazawa signed the SCPA in Tokyo on 14 Mar 2026: early notification of disruptions, an emergency meeting within five days of a request, and prior consultation before measures that could harm the other side; no amounts or named projects.[21]

Investment vehicle, project pipeline and financing

Corporate relationships in the recommended compacts

- LGES agreed to supply Toyota's North American EVs with about 20 GWh a year of modules from 2025, with roughly 4 trillion won (about 3 billion dollars) invested in Michigan lines dedicated to Toyota.[22]
- JERA and Lotte Fine Chemical agreed in Aug 2024 to develop low-carbon hydrogen and ammonia fuel supply chains, with joint studies on commercial frameworks and ammonia bunkering at Yokohama.[23]
- Sumitomo Chemical plans EUV and ArF photoresist production in Korea, previously made only in Osaka, with about 30 billion yen invested across Korea and Japan; site and start date undecided; dollar figure is a rough conversion.[24]

- OCI and Tokuyama hold 50:50 in OCI Tokuyama Semiconductor Materials, which broke ground in Sarawak in July 2025 on an 8,000 tonne a year plant targeting commercial operation in 2029; semiconductor-grade, not solar-grade; amount not stated.[25]
- The 2023 Honda-LGES joint venture L-H Battery keeps operating its Ohio cell plant under a lease after LGES sold the building to Honda for about 4.22 trillion won (2.85 billion dollars) announced Dec 2025; a later report of a lower closing value of 3.74 trillion won is unconfirmed.[26]
- LGES agreed to sell the Ohio JV factory building and related assets to Honda for about $2.85B with the JV continuing via lease and closing targeted for 2026-02-28.[27]
- JERA and Samsung C&T signed a two-year initial MoU on 6 Jul 2026 to explore resilience and supply security in hydrogen and ammonia value chains, including storage and ammonia trading; exploratory, no amounts.[28]
- Samsung's advanced-packaging R&D lab opened in Yokohama's Minato Mirai district in Sept 2026 per a Herald report, after a Dec 2023 plan of about 40 billion yen (280 million dollars); investment figures vary across reports and Japanese materials makers are named partners.[29]
- Unverified items are left out of this section; they stay in the evidence register (evidence_register.csv) and the evidence layer.[1]
- *Inference.* Working channels today: Japan-ROK Hydrogen and Ammonia Cooperation Dialogue and company links (a project pipeline). What the relationship lacks: no cross-sector KPI dashboard or review cycle; no bilateral mechanism covers SC-SOLAR, SC-SEMIS; JPKR-M1 has no published project pipeline.

2. What Each Country Is Good At — and What It Needs

Cell codes: position / need for **Japan · Korea**. Position S strong (top quarter of contenders and above the capability floor), M contender, W not a contender; need H high (supply-chain need >= 0.67, or deployment need >= 0.60 at Deployment), L low; - not measured. Source: strategic_position_tbl (engine choices are in country_profiles.csv).

TECHNOLOGY	EXTRACTION	REFINING	COMPONENTS	ASSEMBLY	DEPLOYMENT	RECOVERY
Batteries	W/H · W/H	M/H · W/H	M/H · M/H	M/H · S/H	S/L · S/L	-/H · -/H
Coal	W/H · W/H	M/H · W/H	S/H · S/L	S/L · S/L	S/L · S/L	-/- · -/-
Electric Grid	W/H · W/H	M/H · W/H	S/L · M/H	M/H · M/L	-/L · -/L	-/- · -/-
Electric Vehicles	W/H · W/H	M/H · W/H	M/L · S/H	M/L · S/L	S/L · S/H	-/- · -/-
Gas	W/H · W/H	W/H · W/H	S/L · M/L	S/L · S/L	S/H · S/L	-/- · -/-
Geothermal	-/- · -/-	-/- · -/-	-/- · -/-	-/- · -/-	S/L · M/H	-/- · -/-
Green Hydrogen	W/- · W/-	W/- · W/-	-/- · -/-	-/- · -/-	-/H · -/H	-/- · -/-
Nuclear	W/L · W/H	W/H · W/H	M/L · S/L	S/L · S/L	S/H · S/H	-/- · -/-
Oil	W/H · W/H	M/L · M/L	M/L · M/H	S/L · S/L	S/L · S/L	-/L · -/L
Solar	W/H · W/H	M/H · W/H	M/L · M/L	M/L · M/L	S/H · M/H	-/- · -/-
Wind	W/- · W/-	W/H · W/H	M/L · M/H	M/H · M/L	S/H · S/H	-/- · -/-

Sources: 1.

Electro-industrial nodes where either country has a position or a need (AIB division of labour, 2024 trade; share = world export or production share; need is the node's technology x stage need, inherited, so a proxy; demand = share of allied demand). 64 of 76 nodes:

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

Table A1 in the appendix (64 rows) carries this table in full; the CSV outputs carry every column.

- Revealed policy priority (NIPO / IEA PAMS within-country percentile; policy never enters the classification) is highest for Japan in: Green Hydrogen assembly (0.99); Green Hydrogen components (0.98); Batteries assembly (0.92); Gas recovery (0.86); Coal assembly (0.84); for Korea in: Batteries assembly (0.96); Green Hydrogen assembly (0.95); Nuclear assembly (0.92); Nuclear components (0.90); Electric Vehicles recovery (0.87).[1]
- Evidence quality: 41 of 66 cells have high stage-specific evidence for both countries; 19 units are flagged low-evidence.[1]

3. Bilateral Complementarity Map

State counts over all units: a_supplies_b_needs 1; b_supplies_a_needs 1; complementary_specialization 13; shared_vulnerability 13; capacity_constrained_trusted_supply 5; direct_competition 6; low_strategic_case 78; deployment_demand 7; not_assessed 18.[1]

Inference. Rules (Section 3 of docs/partnership_memos.md): two strong positions are **direct competition**, never complementarity; a supplier must be strong (or a contender the engine names as the other's trusted partner) and the buyer must have high need, or material demand where trusted capacity is short or a chokepoint exists; **capacity-constrained** where the AIB finds allied capacity short of demand; **shared vulnerability** where both need the node, neither supplies it and a coercive supplier holds at least half of world supply.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

Table A2 in the appendix (39 rows) carries this table in full; the CSV outputs carry every column.

- Low-priority / open-market nodes (55): synthetic graphite (anode-grade), pv modules & generators, li-ion cells & packs, rare-earth metals, uranium conversion & enrichment, fuel assemblies & zirconium, reactor parts, nuclear reactors & enrichment machinery, uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores), rare-earth ores (monazite), electric motors, electric buses, trucks & tractors, ferrite & other magnets, motor & generator parts, electric & plug-in cars, electric two-wheelers & works trucks, wind turbine generating sets, pv cover glass, wind blades & hubs, wind towers, copper smelting & refining, refrigerants, heat pumps & ac units, static converters & inverters, power electronics components, solar thermal water heaters, electrolysers, refrigeration & heat-pump compressors, power semiconductors, sensors & controller parts, insulators & bushings, other accumulators, electricity meters, electric furnaces & heating, furnace parts, lithium minerals, battery recycling feed, transformers, hydro turbine parts, switchgear & protection, graphite electrodes, blade core wood (balsa), ac generators, nickel intermediates & chemicals, copper ores & concentrates, hydrogen, ammonia & methanol, pgms (iridium, ruthenium), nuclear special materials, direct-reduced iron (dri/hbi), cobalt ores, overhead conductors, copper wire & rod, aluminium wire rod, electric rail vehicles, hydro turbines.[1]
- Deployment cells are demand context, not supply relations: high deployment need in Green Hydrogen (JPN and KOR); Nuclear (JPN and KOR); Electric Vehicles (KOR); Solar (JPN and KOR); Wind (JPN and KOR); Geothermal (KOR); Gas (JPN).[1]

4. Trade Agenda

- *Inference.* **What Japan should export to Korea:** Wafers; Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds); Rare-earth permanent magnets; Battery materials (foil, electrolyte); Electric Grid components (insulators & bushings, power electronics components, power semiconductors); Piezo-electric quartz; Gas components; Manganese dioxide (EMD); Grain-oriented electrical steel (GOES); Automatic controllers; Semiconductor manufacturing equipment; Silver powder.
- *Inference.* **What Korea should export to Japan:** Solar cells; Cathode materials & precursors; Nuclear components (fuel assemblies & zirconium, reactor parts); Batteries assembly (li-ion cells & packs, other

accumulators); Gallium, germanium & indium; Non-oriented electrical steel (NGO); Integrated circuits; Power cables.

- *Inference.* **Where tariffs would undermine allied specialisation:** wafers, rare-earth-permanent-magnets.
- *Inference.* **Procurement, standards and content rules to recognise mutually:** Electric Vehicles|Refining.
- *Inference.* **Where third-country circumvention safeguards are genuinely required:** Batteries|Refining, Wind|Refining, rare-earth-oxides-and-compounds, cobalt-intermediates-and-chemicals, lithium-chemicals; elsewhere they would add cost without security.
- *Inference.* **Where unrestricted global sourcing is acceptable:** pgms-iridium-ruthenium, copper-ores-and-concentrates, nickel-intermediates-and-chemicals, nuclear-special-materials, battery-recycling-feed, lithium-minerals.

Table A3 in the appendix (24 rows) carries this table in full; the CSV outputs carry every column.

5. Investment Agenda

Japan → Korea investment (Japan capital and firms expanding in Korea)

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I06	Battery materials (foil, electrolyte)	JPN	KOR	KOR takes 16% of allied demand (0% of allied net imports); AIB class BUILD		anchor offtake	medium	2-3 years
I07	Electric Grid components (insulators & bushings, power electronics components, power semiconductors)	JPN	KOR	supply-chain need 0.35 (JPN) / 0.70 (KOR)	Sumitomo Chemical	market access	medium	2-3 years
I08	Piezo-electric quartz	JPN	KOR	KOR takes 0% of allied demand (0% of allied net imports); AIB class BUILD		export credit	high	2-3 years
I09	Manganese dioxide (EMD)	JPN	KOR	KOR takes 2% of allied demand (2% of allied net imports); AIB class BUILD		standards alignment	high	2-3 years
I10	Grain-oriented electrical steel (GOES)	JPN	KOR	KOR takes 2% of allied demand (0% of allied net imports); AIB class BUILD		standards alignment	high	2-3 years
I11	Automatic controllers	JPN	KOR	KOR takes 6% of allied demand (2% of allied net imports); AIB class LOW INTERVENTION		standards alignment	medium	2-3 years
I12	Semiconductor manufacturing equipment	JPN	KOR	KOR takes 27% of allied demand (16% of allied net imports); AIB class LOW INTERVENTION		standards alignment	medium	2-3 years

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I13	Silver powder	JPN	KOR	KOR takes 16% of allied demand (12% of allied net imports); AIB class LOW INTERVENTION		standards alignment	medium	2-3 years

Sources: 1.

Korea → Japan investment (Korea capital and firms expanding in Japan)

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ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I01	Solar cells	KOR	JPN	JPN takes 2% of allied demand (2% of allied net imports); AIB class BUILD		standards alignment	high	2-3 years
I02	Cathode materials & precursors	KOR	JPN	JPN takes 10% of allied demand (0% of allied net imports); AIB class BUILD		standards alignment	high	2-3 years
I03	Batteries assembly (li-ion cells & packs, other accumulators)	KOR	JPN	supply-chain need 0.71 (JPN) / 0.83 (KOR)	Honda; LG Energy Solution; Toyota	strategic imports	low	2-3 years
I04	Gallium, germanium & indium	KOR	JPN	JPN takes 15% of allied demand (9% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I05	Non-oriented electrical steel (NGO)	KOR	JPN	JPN takes 5% of allied demand (0% of allied net imports); AIB class BUILD		standards alignment	high	2-3 years

Sources: 1.

Joint investment (at home or in third countries)

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I14	Wafers	JPN+KOR	third countries (TWN; USA) and either partner	KOR takes 21% of allied demand (12% of allied net imports); AIB class BUILD		anchor offtake	medium	2-3 years
I15	Rare-earth permanent magnets	JPN+KOR	third countries (PHL; VNM) and either partner	KOR takes 7% of allied demand (6% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years

Sources: 1.

- *Inference.* Additionality = mean of three named components: the market failure (chokepoint 1, capacity constraint 0.75, gap or emerging bottleneck 0.5, none 0), one minus an existing commitment in the evidence layer (committed 1, announced 0.5) and one minus a working market (the buyer already takes 30% or more of

its imports from the supplier). It is **low** whenever there is no failure or the investment is already committed: public support would then not be additional.

6. Third-Country and Development Cooperation

- From the development-partnership pairs (scripts/93: a donor's need meets a developing economy's credible Catch Up pathway in the same step), Japan and Korea are both donors at 10 recipient steps. Ranked: both donors' own need first, then steps inside a recommended compact.[1]
- *Inference.* Roles: Japan brings capital and export finance and specialised technology and components; Korea brings specialised technology and components and equipment and manufacturing at scale (AIB strategic roles).

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	RECIPIENT	TECHNOLOGY	STAGE	COMPACT	RECIPIENT CHOICE	RECIPIENT ANCHOR	JAPAN INTEREST	KOREA INTEREST
X01	Madagascar	Batteries	Refining	SC-BATTERIES	catch up	direct export share; operating capacity	own catch up need (JPN)	own catch up need (KOR)
X02	Philippines	Batteries	Refining	SC-BATTERIES	catch up	direct export share; operating capacity	own catch up need (JPN)	own catch up need (KOR)
X03	Indonesia	Batteries	Refining	SC-BATTERIES	preserve	direct supply share; direct export share; operating capacity; committed pipeline	own catch up need (JPN)	own catch up need (KOR)
X04	Argentina	Batteries	Refining	SC-BATTERIES	catch up	direct supply share; operating capacity; committed pipeline	own catch up need (JPN)	own catch up need (KOR)
X05	India	Batteries	Refining	SC-BATTERIES	catch up	direct export share	own catch up need (JPN)	own catch up need (KOR)
X06	Democratic Republic of Congo	Batteries	Refining	SC-BATTERIES	catch up	direct export share; operating capacity	own catch up need (JPN)	own catch up need (KOR)
X07	Thailand	Batteries	Refining	SC-BATTERIES	catch up	direct export share	own catch up need (JPN)	own catch up need (KOR)
X08	South Africa	Batteries	Refining	SC-BATTERIES	catch up	direct export share	own catch up need (JPN)	own catch up need (KOR)
X09	Malaysia	Batteries	Refining	SC-BATTERIES	catch up	direct export share	own catch up need (JPN)	own catch up need (KOR)
X10	Papua New Guinea	Batteries	Refining	SC-BATTERIES	catch up	direct export share	own catch up need (JPN)	own catch up need (KOR)

Sources: 1.

- DFAT notes RCEP entered into force for Korea on 1 Feb 2022 and may be used by exporters as an alternative to KAFTA; the member list shown here is partial.[2]

- The three leaders agreed to pilot an early warning system exchanging supply-chain information via embassies in selected countries, with critical minerals and rechargeable batteries as priority products, plus an annual commerce and industry ministers track; the system is not published as operating beyond the pilot.[19]
- LGES agreed to supply Toyota's North American EVs with about 20 GWh a year of modules from 2025, with roughly 4 trillion won (about 3 billion dollars) invested in Michigan lines dedicated to Toyota.[22]
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- Senior officials meeting in New York agreed to enhance dialogue on critical minerals and cooperation in Southeast Asia and Sub-Saharan Africa through the trilateral Early Warning System.[6]
- The 2023 Honda-LGES joint venture L-H Battery keeps operating its Ohio cell plant under a lease after LGES sold the building to Honda for about 4.22 trillion won (2.85 billion dollars) announced Dec 2025; a later report of a lower closing value of 3.74 trillion won is unconfirmed.[26]
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- The US-hosted ministerial in Feb 2026 was attended by Japan and Korea among others, and Korea was reported to chair the FORGE critical-minerals forum through June 2026.[10]
- Korea chaired the Minerals Security Partnership until June 2026; a February 2026 report says its 17 members carried into the successor FORGE; Korean public lenders joined the MSP Finance Network; no signed US-Korea critical-minerals MOU was found.[11]

7. Recommended Sectoral Compacts

Inference. 5 compacts pass (rule: at least one joint bet, at least two active nodes and both countries holding an AIB compact role; or, labelled **evidence-led**, an existing bilateral pipeline or at least 2 bilateral project items where OSI measures a need but cannot see the partner's capability). The hypothesis test for every AIB compact:

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

RANK	COMPACT	VERDICT	JOINT BETS	COORDINATE	JAPAN ROLE	KOREA ROLE	BILATERAL MECHANISMS	RECOMMENDED
1	Battery Materials & Cells	bilateral priority	4	2	Processing anchor	Specialized technology supplier	JPKR-M2; JPKR-M6	TRUE
2	Permanent Magnets & Drive Systems	bilateral priority	2	1	Scale manufacturer	Specialized technology supplier	JPKR-M2; JPKR-M6	TRUE
3	Solar Manufacturing	bilateral priority	1	3	Specialized technology supplier	Specialized technology supplier		TRUE
4	Semiconductors & Power Electronics	evidence-led candidate	0	5	Specialized technology supplier	Specialized technology supplier	JPKR-M5	TRUE
5	Industrial Electrification & Hydrogen	evidence-led candidate	0	0	Scale manufacturer	Scale manufacturer	JPKR-M3	TRUE
6	Grid Equipment & Conductors	secondary	0	2	Specialized technology supplier	Scale manufacturer		FALSE

RANK	COMPACT	VERDICT	JOINT BETS	COORDINATE	JAPAN ROLE	KOREA ROLE	BILATERAL MECHANISMS	RECOMMENDED
7	Electric Vehicles	not a bilateral priority	0	0	Scale manufacturer	Scale manufacturer		FALSE
8	Heat Pumps & Cooling	not a bilateral priority	0	0	Specialized technology supplier	Scale manufacturer		FALSE
9	Nuclear Fuel & Reactor Systems	not a bilateral priority	0	0	Capacity-building host	Scale manufacturer		FALSE
10	Wind & Hydropower Equipment	not a bilateral priority	0	0	Bridge / coordination partner	Specialized technology supplier		FALSE

Sources: 1.

1. Battery Materials & Cells

Strategic objective

- AIB mission: Build missing capacity in Li-ion cells & packs and lithium chemicals and manganese dioxide (EMD) (refined chemicals); targeted: preserve and second-source specialized production in lithium chemicals (refined chemicals)..[1]
- *Recommendation.* Change the outcome at natural graphite, cobalt intermediates & chemicals, lithium chemicals: geopolitical chokepoint.

Why this pair?

- *Inference.* Natural graphite: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 82% of world supply). Cobalt intermediates & chemicals: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 76% of world supply). Lithium chemicals: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 74% of world supply). Battery materials (foil, electrolyte): Capacity-constrained trusted supply (JPN is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Cathode materials & precursors: Complementary specialisation (KOR is strong here and the other country is strong at another step of the same chain). Manganese dioxide (EMD): Complementary specialisation (JPN is strong here and the other country is strong at another step of the same chain).
- AIB compact roles: Japan Processing anchor (lead node manganese-dioxide-emd); Korea Specialized technology supplier (lead node cathode-materials-and-precursors).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	JAPAN POSITION	KOREA POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	natural-graphite	AIB node	moderate	moderate	shared vulnerability	BUILD	anchor offtake
Refining	cobalt-intermediates-and-chemicals	AIB node	moderate	moderate	shared vulnerability	BUILD	anchor offtake
Components	accumulator-parts-and-separators	AIB node	strong	strong	direct competition	MANAGE SPECIALIZATION	standards alignment

STAGE	UNIT	BASIS	JAPAN POSITION	KOREA POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Assembly	li-ion-cells-and-packs	AIB node	moderate	moderate	low strategic case	LOW INTERVENTION	
Deployment	Batteries/Deployment	OSI cell (no AIB node at this stage)	strong	strong	low strategic case	LOW INTERVENTION	
Recovery	battery-recycling-feed	AIB node	moderate	weak	low strategic case	LOW INTERVENTION	

Sources: 1.

Reciprocal bargain

- Japan provides: supply of battery materials (foil, electrolyte), manganese dioxide (emd); long-term demand and offtake for cathode materials & precursors; procurement recognition of the partner as trusted supply; finance for the compact's nodes (JBIC, NEXI, JOGMEC).
- Korea provides: capacity and technology in cathode materials & precursors; long-term demand for battery materials (foil, electrolyte), manganese dioxide (emd); procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KEXIM, K-SURE, KOMIR, KDB).
- Jointly: pooled offtake and finance for non-coercive natural graphite, cobalt intermediates & chemicals, lithium chemicals.
- Third-country partners needed: MDG, MOZ, DEU, COD, FIN, ARG, CHL, MYS, TWN, GRC, ESP.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): MDG at natural graphite; FIN at cobalt intermediates & chemicals; CHL at lithium chemicals; DEU at synthetic graphite (anode-grade); POL at li-ion cells & packs; VNM at other accumulators. The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* anchor_offtake for natural graphite; anchor_offtake for cobalt intermediates & chemicals; anchor_offtake for lithium chemicals; anchor_offtake for battery materials (foil, electrolyte); standards_alignment for cathode materials & precursors; standards_alignment for manganese dioxide (emd)

Priority transactions

- *Recommendation.* Long-term offtake by Japan and Korea buyers for non-coercive electric vehicles extraction (cobalt ores, lithium minerals, natural graphite), sized to underwrite new capacity (candidate hosts: CHL, ARG, BRA) (D02).

Success by 2030

- Natural graphite: coercive share of supply to the two buyers below 50% (CHN now holds 82% of world supply).
- Cobalt intermediates & chemicals: coercive share of supply to the two buyers below 50% (CHN now holds 76% of world supply).
- Lithium chemicals: coercive share of supply to the two buyers below 50% (CHN now holds 74% of world supply).
- Battery materials (foil, electrolyte): allied residual gap under the China shock closed (now 14% of allied demand); Japan's share of Korea's imports up from 11%.
- Cathode materials & precursors: coercive share of supply to the two buyers below 50% (CHN now holds 92% of world supply); Korea's share of Japan's imports up from 3%.
- Manganese dioxide (EMD): allied residual gap under the China shock closed (now 34% of allied demand); Japan's share of Korea's imports up from 11%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- The three leaders agreed to pilot an early warning system exchanging supply-chain information via embassies in selected countries, with critical minerals and rechargeable batteries as priority products, plus an annual commerce and industry ministers track; the system is not published as operating beyond the pilot.[19]
- LGES agreed to supply Toyota's North American EVs with about 20 GWh a year of modules from 2025, with roughly 4 trillion won (about 3 billion dollars) invested in Michigan lines dedicated to Toyota.[22]
- A 2024 report says LGES sources separators from China's SEMCORP and Senior and from Japan's Asahi Kasei and Toray, while Samsung SDI relies mainly on Korea's WCP; no Korea-based Japanese separator plant was found in 2025-26 coverage.[30]
- Korean and Japanese officials agreed a close cooperation system on critical-mineral supply risks, support for company and agency cooperation on joint development in third countries (Latin America, Asia, Africa), and policy coordination in the Minerals Security Partnership and IEA; no projects were named.[4]
- Senior officials meeting in New York agreed to enhance dialogue on critical minerals and cooperation in Southeast Asia and Sub-Saharan Africa through the trilateral Early Warning System.[6]
- The second round discussed a joint response to falling copper treatment and refining charges, cooperation within the Minerals Security Partnership and ways to develop critical minerals jointly in third countries; no agreement or named project was reported.[8]
- The 2023 Honda-LGES joint venture L-H Battery keeps operating its Ohio cell plant under a lease after LGES sold the building to Honda for about 4.22 trillion won (2.85 billion dollars) announced Dec 2025; a later report of a lower closing value of 3.74 trillion won is unconfirmed.[26]
- LGES agreed to sell the Ohio JV factory building and related assets to Honda for about $2.85B with the JV continuing via lease and closing targeted for 2026-02-28.[27]

2. Permanent Magnets & Drive Systems

Strategic objective

- AIB mission: Build missing capacity in rare-earth permanent magnets (magnets & core steel)..[1]
- *Recommendation.* Change the outcome at rare-earth oxides & compounds, rare-earth permanent magnets, non-oriented electrical steel (ngo): geopolitical chokepoint; trusted partner capacity constraint.

Why this pair?

- *Inference.* Rare-earth oxides & compounds: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 84% of world supply). Rare-earth permanent magnets: Capacity-constrained trusted supply (JPN is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Non-oriented electrical steel (NGO): Complementary specialisation (KOR is strong here and the other country is strong at another step of the same chain).
- AIB compact roles: Japan Scale manufacturer (lead node non-oriented-electrical-steel-ngo); Korea Specialized technology supplier (lead node non-oriented-electrical-steel-ngo).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	JAPAN POSITION	KOREA POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	rare-earth-ores-monazite	AIB node	weak	weak	low strategic case	LOW INTERVENTION	
Refining	rare-earth-oxides-and-compounds	AIB node	moderate	weak	shared vulnerability	BUILD	anchor offtake
Components	rare-earth-permanent-magnets	AIB node	strong	weak	capacity constrained trusted supply	BUILD	anchor offtake
Assembly	electric-motors	AIB node	moderate	moderate	low strategic case	LOW INTERVENTION	
Deployment	Electric Vehicles/Deployment	OSI cell (no AIB node at this stage)	strong	strong	deployment demand	DEMAND CONTEXT	
Recovery	Electric Vehicles/Recovery	OSI cell (no AIB node at this stage)	not measured	not measured	not assessed	INSUFFICIENT DATA	

Sources: 1.

Reciprocal bargain

- Japan provides: supply of rare-earth permanent magnets; long-term demand and offtake for non-oriented electrical steel (ngo); procurement recognition of the partner as trusted supply; finance for the compact's nodes (JBIC, NEXI, JOGMEC).
- Korea provides: capacity and technology in non-oriented electrical steel (ngo); long-term demand for rare-earth permanent magnets; procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KEXIM, K-SURE, KOMIR, KDB).
- Jointly: pooled offtake and finance for non-coercive rare-earth oxides & compounds.
- Third-country partners needed: USA, MYS, PHL, VNM, TWN.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): MYS at rare-earth oxides & compounds; VNM at rare-earth metals; DEU at electric motors; DEU at ferrite & other magnets; DEU at motor & generator parts. The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* anchor_offtake for rare-earth oxides & compounds; anchor_offtake for rare-earth permanent magnets; standards_alignment for non-oriented electrical steel (ngo)

Priority transactions

- *Recommendation.* Long-term offtake by Korea buyers for Japan rare-earth permanent magnets, sized to underwrite new capacity (candidate hosts: PHL, VNM) (D01).

Success by 2030

- Rare-earth oxides & compounds: coercive share of supply to the two buyers below 50% (CHN now holds 84% of world supply).
- Rare-earth permanent magnets: coercive share of supply to the two buyers below 50% (CHN now holds 64% of world supply); Japan's share of Korea's imports up from 2%.
- Non-oriented electrical steel (NGO): allied residual gap under the China shock closed (now 17% of allied demand); Korea's share of Japan's imports up from 2%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- A 2023 KED Global analysis put Korea's dependence on Chinese neodymium permanent magnets at 88% versus 31% for Japan; no Japanese magnet-maker plant or supply deal for Korea was found in this research.[31]
- On 13 Jan 2026 the leaders discussed supply-chain cooperation in depth and agreed to launch agency-level talks on economic security cooperation, with rare-earth stabilisation reported as a focus; no outputs were published.[9]
- The US-hosted ministerial in Feb 2026 was attended by Japan and Korea among others, and Korea was reported to chair the FORGE critical-minerals forum through June 2026.[10]

3. Solar Manufacturing

Strategic objective

- AIB mission: Build missing capacity in modules, cells & cover glass and polysilicon..[1]
- *Recommendation.* Change the outcome at wafers, polysilicon, solar cells: geopolitical chokepoint.

Why this pair?

- *Inference.* Wafers: Capacity-constrained trusted supply (JPN is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Polysilicon: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 97% of world supply). Solar cells: Complementary specialisation (KOR is strong here and the other country is strong at another step of the same chain). Silver powder: Complementary specialisation (JPN is strong here and the other country is strong at another step of the same chain).
- AIB compact roles: Japan Specialized technology supplier (lead node silver-powder); Korea Specialized technology supplier (lead node solar-cells).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	JAPAN POSITION	KOREA POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	Solar/Extraction	OSI cell (no AIB node at this stage)	weak	weak	low strategic case	LOW INTERVENTION	
Refining	polysilicon	AIB node	moderate	moderate	shared vulnerability	BUILD	anchor offtake
Components	solar-cells	AIB node	moderate	strong	complementary specialization	COORDINATE	standards alignment
Assembly	pv-modules-and-generators	AIB node	weak	weak	low strategic case	LOW INTERVENTION	
Deployment	Solar/Deployment	OSI cell (no AIB node at this stage)	strong	moderate	deployment demand	DEMAND CONTEXT	
Recovery	Solar/Recovery	OSI cell (no AIB node at this stage)	not measured	not measured	not assessed	INSUFFICIENT DATA	

Sources: 1.

Reciprocal bargain

- Japan provides: supply of wafers, silver powder; long-term demand and offtake for solar cells; procurement recognition of the partner as trusted supply; finance for the compact's nodes (JBIC, NEXI, JOGMEC).
- Korea provides: capacity and technology in solar cells; long-term demand for wafers, silver powder; procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KEXIM, K-SURE, KOMIR, KDB).
- Jointly: pooled offtake and finance for non-coercive polysilicon.
- Third-country partners needed: TWN, USA, MYS, DEU, THA.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): VNM at pv modules & generators; MYS at pv cover glass. The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* anchor_offtake for wafers; anchor_offtake for polysilicon; standards_alignment for solar cells; standards_alignment for silver powder

Priority transactions

- *Recommendation.* Long-term offtake by Korea buyers for Japan wafers, sized to underwrite new capacity (candidate hosts: TWN, USA) (D03).

Success by 2030

- Wafers: coercive share of supply to the two buyers below 50% (CHN now holds 98% of world supply); Japan's share of Korea's imports up from 15%.
- Polysilicon: coercive share of supply to the two buyers below 50% (CHN now holds 97% of world supply).
- Solar cells: coercive share of supply to the two buyers below 50% (CHN now holds 92% of world supply); Korea's share of Japan's imports up from 13%.
- Silver powder: allied residual gap under the China shock closed (now 0% of allied demand); Japan's share of Korea's imports up from n/a.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- OCI and Tokuyama hold 50:50 in OCI Tokuyama Semiconductor Materials, which broke ground in Sarawak in July 2025 on an 8,000 tonne a year plant targeting commercial operation in 2029; semiconductor-grade, not solar-grade; amount not stated.[25]

4. Semiconductors & Power Electronics (evidence-led)

- *Inference.* **Basis: external evidence.** OSI trade data show no bilateral complementarity at this compact's nodes (Korea world share: piezo-electric quartz 0.0%, gallium, germanium & indium 18.0%, automatic controllers 2.8%, integrated circuits 17.1%, semiconductor manufacturing equipment 5.6%, static converters & inverters 1.3%, power electronics components 2.5%, power semiconductors 3.2%, sensors & controller parts 2.4%); the case rests on the bilateral pipeline and corporate evidence listed at the end of this compact.

- Need that OSI does measure in the compact: piezo-electric quartz (Japan demand n/a, 19% of allied demand; Korea not_measured n/a); gallium, germanium & indium (Japan demand n/a, 15% of allied demand; Korea low n/a); automatic controllers (Japan low (cell 0.67; not a material buyer), 6% of allied demand; Korea demand 0.51); semiconductor manufacturing equipment (Japan low 0.55, 8% of allied demand; Korea demand 0.25); sensors & controller parts (Japan demand n/a, 12% of allied demand; Korea low n/a).[1]

Strategic objective

- AIB mission: Monitor: no intervention class reaches the materiality floor (25% of the compact's supply to shift or US$1bn); targeted: build missing capacity in piezo-electric quartz (semiconductor materials); preserve and second-source specialized production in piezo-electric quartz (semiconductor materials)..[1]
- *Recommendation.* Change the outcome at piezo-electric quartz, gallium, germanium & indium, automatic controllers: geopolitical chokepoint; trusted partner capacity constraint; strategic manufacturing gap.

Why this pair?

- *Inference.* Piezo-electric quartz: Complementary specialisation (JPN is strong here and the other country is strong at another step of the same chain). Gallium, germanium & indium: Capacity-constrained trusted supply (KOR is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Automatic controllers: Complementary specialisation (JPN is strong here and the other country is strong at another step of the same chain). Integrated circuits: Complementary specialisation (KOR is strong here and the other country is strong at another step of the same chain). Semiconductor manufacturing equipment: Complementary specialisation (JPN is strong here and the other country is strong at another step of the same chain).
- AIB compact roles: Japan Specialized technology supplier (lead node semiconductor-manufacturing-equipment); Korea Specialized technology supplier (lead node integrated-circuits).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	JAPAN POSITION	KOREA POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	Electric Grid/Extraction	OSI cell (no AIB node at this stage)	weak	weak	low strategic case	LOW INTERVENTION	
Refining	piezo-electric-quartz	AIB node	strong	weak	complementary specialization	COORDINATE	standards alignment
Components	integrated-circuits	AIB node	moderate	strong	complementary specialization	COORDINATE	standards alignment
Assembly	automatic-controllers	AIB node	strong	moderate	complementary specialization	COORDINATE	standards alignment
Deployment	Electric Grid/Deployment	OSI cell (no AIB node at this stage)	not measured	not measured	low strategic case	LOW INTERVENTION	
Recovery	Electric Grid/Recovery	OSI cell (no AIB node at this stage)	not measured	not measured	not assessed	INSUFFICIENT DATA	

Sources: 1.

Reciprocal bargain

- Japan provides: supply of piezo-electric quartz, automatic controllers, semiconductor manufacturing equipment; long-term demand and offtake for gallium, germanium & indium, integrated circuits; procurement recognition of the partner as trusted supply; finance for the compact's nodes (JBIC, NEXI, JOGMEC).

- Korea provides: capacity and technology in gallium, germanium & indium, integrated circuits; long-term demand for piezo-electric quartz, automatic controllers, semiconductor manufacturing equipment; procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KEXIM, K-SURE, KOMIR, KDB).
- Jointly: common specifications and qualification.
- Third-country partners needed: TWN, PHL, SGP, USA, DEU, MYS.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): DEU at static converters & inverters; DEU at power electronics components; MYS at power semiconductors; USA at sensors & controller parts. The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* standards_alignment for piezo-electric quartz; anchor_offtake for gallium, germanium & indium; standards_alignment for automatic controllers; standards_alignment for integrated circuits; standards_alignment for semiconductor manufacturing equipment

Priority transactions

- *Recommendation.* Tariff binding for Japan electric grid components (insulators & bushings, power electronics components, power semiconductors) entering Korea (D04).
- *Recommendation.* Long-term offtake by Japan buyers for Korea gallium, germanium & indium, sized to underwrite new capacity (candidate hosts: SGP, USA) (D07).

Success by 2030

- Piezo-electric quartz: coercive share of supply to the two buyers below 50% (CHN now holds 60% of world supply); Japan's share of Korea's imports up from n/a.
- Gallium, germanium & indium: allied residual gap under the China shock closed (now 29% of allied demand); Korea's share of Japan's imports up from 3%.
- Automatic controllers: allied residual gap under the China shock closed (now 8% of allied demand); Japan's share of Korea's imports up from 10%.
- Integrated circuits: allied residual gap under the China shock closed (now 6% of allied demand); Korea's share of Japan's imports up from 17%.
- Semiconductor manufacturing equipment: allied residual gap under the China shock closed (now 0% of allied demand); Japan's share of Korea's imports up from 12%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- From July 2019 Japan required individual licences for fluorinated polyimide, photoresist and hydrogen fluoride exports to Korea and removed Korea from its white list; Korea removed Japan from its own list in Aug 2019 and filed a WTO complaint; Japan lifted the three-item curbs in Mar 2023.[17]
- Press coverage of the 2019 dispute put Japan's share of the three controlled materials used by Korean chipmakers at about 90% for fluorinated polyimide and photoresist and about 70% for hydrogen fluoride.[32]

- Korea completed procedures to drop its 2019 WTO complaint on 23 Mar 2023, a week after the 16 Mar Yoon-Kishida summit; Japan had pledged to lift the three-material curbs; the article does not confirm the curbs had by then been lifted.[18]
- Sumitomo Chemical plans EUV and ArF photoresist production in Korea, previously made only in Osaka, with about 30 billion yen invested across Korea and Japan; site and start date undecided; dollar figure is a rough conversion.[24]
- Samsung's advanced-packaging R&D lab opened in Yokohama's Minato Mirai district in Sept 2026 per a Herald report, after a Dec 2023 plan of about 40 billion yen (280 million dollars); investment figures vary across reports and Japanese materials makers are named partners.[29]

5. Industrial Electrification & Hydrogen (evidence-led)

- *Inference.* **Basis: external evidence.** OSI trade data show no bilateral complementarity at this compact's nodes (Korea world share: electrolysers 5.7%, electric furnaces & heating 5.3%, furnace parts 2.2%, graphite electrodes 0.0%, hydrogen, ammonia & methanol 0.2%, pgms (iridium, ruthenium) 0.0%, direct-reduced iron (dri/hbi) 0.0%); the case rests on the bilateral pipeline and corporate evidence listed at the end of this compact.
- Need that OSI does measure in the compact: graphite electrodes (Japan low n/a, 4% of allied demand; Korea demand n/a); hydrogen, ammonia & methanol (Japan demand n/a, 4% of allied demand; Korea demand n/a); pgms (iridium, ruthenium) (Japan demand n/a, 49% of allied demand; Korea not_measured n/a); direct-reduced iron (dri/hbi) (Japan not_measured n/a, 0% of allied demand; Korea demand n/a).[1]

Strategic objective

- AIB mission: Build missing capacity in catalyst metals; preserve and second-source specialized production in catalyst metals..[1]
- *Recommendation.* Use JPKR-M3 to tie the announced projects to the nodes where OSI measures need, and verify the partner capability that trade data cannot see (ownership abroad, domestic sales, heavy components) before any public finance is added.

Why this pair?

- *Inference.* No OSI-observed complementarity; see the evidence below.
- AIB compact roles: Japan Scale manufacturer (lead node electrolysers); Korea Scale manufacturer (lead node electrolysers).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	JAPAN POSITION	KOREA POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction		not covered				no intervention	
Refining	hydrogen-ammonia-and-methanol	AIB node	weak	weak	low strategic case	LOW INTERVENTION	
Components	furnace-parts	AIB node	strong	moderate	low strategic case	LOW INTERVENTION	
Assembly	electrolysers	AIB node	strong	moderate	low strategic case	LOW INTERVENTION	
Deployment		not covered				no intervention	

STAGE	UNIT	BASIS	JAPAN POSITION	KOREA POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Recovery		not covered				no intervention	

Sources: 1.

Reciprocal bargain

- Japan provides: procurement recognition of the partner as trusted supply; finance for the compact's nodes (JBIC, NEXI, JOGMEC).
- Korea provides: procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KEXIM, K-SURE, KOMIR, KDB).
- Jointly: common specifications and qualification.
- Third-country partners needed: none identified by the AIB host model.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): DEU at electric furnaces & heating; ESP at graphite electrodes. The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* Use the existing investment vehicle; no new instrument.

Priority transactions

- See Section 9; no OSI-ranked deal in this compact.

Success by 2030

- Projects in the pipeline reach final investment decision with partner-made equipment in the bill of materials.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- JERA and Lotte Fine Chemical agreed in Aug 2024 to develop low-carbon hydrogen and ammonia fuel supply chains, with joint studies on commercial frameworks and ammonia bunkering at Yokohama.[23]
- KNOC reportedly signed an MOU with JOGMEC in February 2025 on cooperation in hydrogen, ammonia and CCS.[20]
- The dialogue, set up Feb 2024 with a first meeting in Jun 2024, held its second meeting in Tokyo on 26 Mar 2025 and agreed to develop a Japan-ROK cooperation platform for hydrogen and ammonia supply chains; the METI page could not be opened.[5]
- JERA and Samsung C&T signed a two-year initial MoU on 6 Jul 2026 to explore resilience and supply security in hydrogen and ammonia value chains, including storage and ammonia trading; exploratory, no amounts.[28]

8. Master Energy Agreement Assessment

- *Inference.* Rule: **existing** if every recommended compact is covered by a mechanism with both a pipeline and finance and a cross-sector mechanism has KPIs and a review cycle; **upgrade** if at least 50% of compacts are

covered by a mechanism with finance or a pipeline, or a cross-sector mechanism with a pipeline exists; **new MEA** otherwise.

- *Inference.* Result: **upgrade**; coverage 20% of recommended compacts; cross-sector mechanism with KPIs and review: no.

COMPACT	ANY MECHANISM	FINANCE OR PIPELINE	PIPELINE AND FINANCE
SC-BATTERIES	TRUE	FALSE	FALSE
SC-MAGNETS	TRUE	FALSE	FALSE
SC-SOLAR	FALSE	FALSE	FALSE
SC-SEMIS	FALSE	FALSE	FALSE
SC-INDUSTRY	TRUE	TRUE	FALSE

Sources: 1.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	MECHANISM	STATUS	SCOPE	PIPELINE	FINANCE	KPIS	REVIEW	RECIPROCAL
JPKR-M1	Korea-Japan Supply Chain Partnership Arrangement and Industrial and Trade Policy Dialogue	signed	cross sector	FALSE	FALSE	FALSE	partial	TRUE
JPKR-M2	Korea-Japan Mineral Resources Cooperation Dialogue	operating	SC-BATTERIES/SC-MAGNETS	FALSE	FALSE	FALSE	partial	TRUE
JPKR-M3	Japan-ROK Hydrogen and Ammonia Cooperation Dialogue and company links	operating	SC-INDUSTRY	partial	FALSE	FALSE	partial	TRUE
JPKR-M4	RCEP and Korea's CPTPP accession question	in force	cross sector	FALSE	FALSE	FALSE	partial	TRUE
JPKR-M5	2019 export-control dispute and its 2023 settlement	cancelled	SC-SEMIS	FALSE	FALSE	FALSE	FALSE	TRUE
JPKR-M6	US-Japan-ROK trilateral supply-chain early warning and critical-minerals dialogue	announced	SC-BATTERIES/SC-MAGNETS	FALSE	FALSE	FALSE	partial	TRUE
JPKR-M7	FKI-Keidanren business conferences	operating	cross sector	FALSE	FALSE	FALSE	TRUE	TRUE

Sources: 1.

Recommended: upgrade Korea-Japan Supply Chain Partnership Arrangement and Industrial and Trade Policy Dialogue into the execution framework for the compacts

- **Scope:** Battery Materials & Cells; Permanent Magnets & Drive Systems; Solar Manufacturing; Semiconductors & Power Electronics; Industrial Electrification & Hydrogen; no other sectors.
- **Reciprocal commitments:** Japan commits demand certainty (procurement recognition and anchor offtake) and market-access predictability for the nodes Korea supplies (solar cells, cathode materials & precursors, gallium, germanium & indium, non-oriented electrical steel (ngo)); Korea commits the same for the nodes Japan supplies (wafers, rare-earth permanent magnets, battery materials (foil, electrolyte), piezo-electric quartz). Each commits public finance for the compact nodes, at home, in the partner or with named third-country hosts.
- **Compact portfolio:** Battery Materials & Cells (bilateral priority); Permanent Magnets & Drive Systems (bilateral priority); Solar Manufacturing (bilateral priority); Semiconductors & Power Electronics (evidence-led candidate); Industrial Electrification & Hydrogen (evidence-led candidate).

- **Financing lanes:** Japan: JBIC, NEXI, JOGMEC; Korea: KEXIM, K-SURE, KOMIR, KDB; demand-side tools (offtake, stockpiles, procurement): METI, JOGMEC and MCEE, KOMIR, PPS, NSSC; third-country finance: JICA and KEXIM, KOMIR, KOICA.
- **Governance:** one bilateral steering line inside Korea-Japan Supply Chain Partnership Arrangement and Industrial and Trade Policy Dialogue, with the existing sector mechanisms running their workstreams (Japan-ROK Hydrogen and Ammonia Cooperation Dialogue and company links). Projects are selected by a joint committee, not by one side.
- **Milestones:** the 12-month milestones of the deal pipeline (Section 9).
- **KPI dashboard:** per compact node, the coercive share of each country's supply, the bilateral import share, offtake signed (volume), capacity at FID and operating, and public money committed vs private.
- **Review cycle:** every 12 months, with the OSI tables re-run on new trade data.
- **Sunset / exit:** each compact carries the exit conditions in Section 7; the framework expires with the investment vehicle () unless renewed on evidence.

9. 12–36 Month Deal Pipeline

Inference. Ranked lexicographically on separate fields (importance, then additionality, urgency, tractability, execution speed, value-chain interdependency, evidence quality); urgency is not importance. At most two deals per compact.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

RANK	DEAL / ACTION	COMPACT	INSTRUMENT	RESPONSIBLE INSTITUTIONS	12M MILESTONE	IMPORTANCE	URGENCY	ADDITIONALITY
1	Long-term offtake by Korea buyers for Japan rare-earth permanent magnets, sized to underwrite new capacity (candidate hosts: PHL, VNM)	SC-MAGNETS	anchor offtake	JPN: METI, JOGMEC; KOR: MCEE, PPS	Signed long-term offtake from at least one large buyer	high	high	high
2	Long-term offtake by Japan and Korea buyers for non-coercive electric vehicles extraction (cobalt ores, lithium minerals, natural graphite), sized to underwrite new capacity (candidate hosts: CHL, ARG, BRA)	SC-BATTERIES	anchor offtake	JPN: METI, JOGMEC; KOR: MCEE, PPS	Signed long-term offtake from at least one large buyer	high	high	high
3	Long-term offtake by Korea buyers for Japan wafers, sized to underwrite new capacity (candidate hosts: TWN, USA)	SC-SOLAR	anchor offtake	JPN: METI, JOGMEC; KOR: MCEE, PPS	Signed long-term offtake from at least one large buyer	high	high	medium
4	Tariff binding for Japan electric grid components (insulators & bushings, power electronics components, power semiconductors) entering Korea	SC-SEMIS	market access	JPN: METI; KOR: MOTIR (MOTIE), K-SURE, KOTRA	Tariff treatment of the node bound or exempted for the trusted supplier	high	low	medium
5	Export-credit and investment-insurance cover for coal refining transactions		export credit	JPN: JBIC, NEXI, JOGMEC; KOR: KEXIM, K-SURE, KOMIR, KDB	Cover offered for named transactions	high	low	medium
6	Long-term offtake by Japan and Korea buyers for non-coercive solar refining (copper wire & rod, polysilicon), sized to underwrite new capacity (candidate hosts: MYS, DEU)	SC-GRID	anchor offtake	JPN: METI, JOGMEC; KOR: MCEE, PPS	Signed long-term offtake from at least one large buyer	high	high	low

RANK	DEAL / ACTION	COMPACT	INSTRUMENT	RESPONSIBLE INSTITUTIONS	12M MILESTONE	IMPORTANCE	URGENCY	ADDITIONALITY
7	Long-term offtake by Japan buyers for Korea gallium, germanium & indium, sized to underwrite new capacity (candidate hosts: SGP, USA)	SC-SEMIS	anchor offtake	JPN: METI, JOGMEC; KOR: MCEE, PPS	Signed long-term offtake from at least one large buyer	medium	low	high
8	Joint specification work plan for grain-oriented electrical steel (goes)	SC-GRID	standards alignment	JPN: METI; KOR: MOTIR (MOTIE), MCEE, NSSC	Joint list of specifications that fragment the market and a work plan to align them	low	low	high

Sources: 1.

10. What Not to Do

- *Recommendation.* **N01** Do not co-subsidise new Coal components capacity: both are strong suppliers (Japan 9%, Korea 6%). Trusted supply is adequate: this is commercial competition.
- *Recommendation.* **N02** Do not co-subsidise new Accumulator parts & separators capacity: both are strong suppliers (Japan 11%, Korea 13%). More trusted capacity is needed, so agree who specialises in what rather than racing.
- *Recommendation.* **N03** Do not co-subsidise new Coal assembly capacity: both are strong suppliers (Japan 5%, Korea 4%). Trusted supply is adequate: this is commercial competition.
- *Recommendation.* **N04** Do not create a new bilateral energy MOU or a free-standing MEA institution: JPKR-M1 already exists (a standing framework with no pipeline or finance yet); dock the compacts in it.
- *Recommendation.* **N05** Do not put new public money behind Batteries assembly (li-ion cells & packs, other accumulators) investment that is already committed (evidence UJ-037; JPKR-29; JPKR-30); support is additional only beyond it.
- *Recommendation.* **N06** Do not put new public money behind Integrated circuits investment that is already committed (evidence JPKR-05; JPKR-06; JPKR-07; JPKR-31; JPKR-32); support is additional only beyond it.

11. Evidence Register

- Every recommendation (IDs J, C, T, I, X, D, N, M) has rows in evidence_register.csv: the OSI metrics behind it (variable, value for each country, year, source, evidence specificity) and any external items, with URL and access date. Recommendations that rest on external research only carry basis = external.[1]
- External evidence vintage: accessed 2026-11-06; 256 items (partial 197, unverified 18, verified 41).[1]

Where the findings are weak or counterintuitive

- Joint bets resting on a node need inherited from its cell (a proxy): Wafers; Rare-earth oxides & compounds; Natural graphite.
- Evidence-led compacts (OSI cannot see the partner's capability in trade data): Semiconductors & Power Electronics; Industrial Electrification & Hydrogen.
- Korea has no listed trade position (under 1% of world supply) at nodes where Japan has material demand: rare-earth oxides & compounds, rare-earth permanent magnets, cobalt intermediates & chemicals, lithium chemicals, piezo-electric quartz, pv modules & generators, rare-earth metals, rare-earth ores (monazite), wind turbine generating sets, refrigerants, lithium minerals, hydro turbine parts, copper ores & concentrates,

hydrogen, ammonia & methanol, pgms (iridium, ruthenium). Trade data miss capability that is owned abroad or sold at home, so these nodes cannot show complementarity even where corporate evidence suggests it.

- Low-evidence units among the active states: none.
- Fuel-trade lanes reach tier 1 (coal refining; buyer: both): real energy-security lanes, but outside the electro-industrial compacts and with no market failure beyond the supply concentration the engine measures.

Sources

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Appendix — evidence tables and method

The tables below carry the full complementarity map, the node profiles and the trade actions that the main text summarises. Method, decision rules and limitations: docs/partnership_memos.md. Machine-readable outputs: country_profiles.csv, dyad_complementarity.csv, compact_candidates.csv, trade_priorities.csv, investment_priorities.csv, third_country_opportunities.csv, deal_pipeline.csv and evidence_register.csv.

Table A1. Section 2, What Each Country Is Good At — and What It Needs

COMPACT	NODE	STAGE	JAPAN SHARE	JAPAN NEED	KOREA SHARE	KOREA NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-BATTERIES	Natural graphite	Extraction	1.7%	high 0.98	2.6%	high 1.00	BUILD	CHN 82%
SC-BATTERIES	Lithium minerals	Extraction	0.4%	high 0.98	0.3%	high 1.00	LOW INTERVENTION	CHN 25%
SC-BATTERIES	Battery recycling feed	Recovery	1.6%	low (cell 0.98; not a material buyer)	0.3%	high 0.94	LOW INTERVENTION	CHN 8%
SC-BATTERIES	Cobalt intermediates & chemicals	Refining	2.4%	high 0.94	0.8%	high 0.99	BUILD	CHN 76%
SC-BATTERIES	Lithium chemicals	Refining	0.1%	high 0.94	0.7%	high 0.99	BUILD	CHN 74%
SC-BATTERIES	Manganese dioxide (EMD)	Refining	18.3%	low (cell 0.94; not a material buyer)	0.0%	low (cell 0.99; not a material buyer)	BUILD	CHN 37%
SC-BATTERIES	Nickel intermediates & chemicals	Refining	2.9%	high 0.94	1.0%	low (cell 0.99; not a material buyer)	LOW INTERVENTION	CHN 37%
SC-BATTERIES	Battery materials (foil, electrolyte)	Refining	11.2%	low (cell 0.94; not a material buyer)	9.5%	high 0.99	BUILD	CHN 16%
SC-BATTERIES	Cathode materials & precursors	Refining	9.7%	low (cell 0.94; not a material buyer)	50.9%	low (cell 0.99; not a material buyer)	BUILD	CHN 92%
SC-BATTERIES	Accumulator parts & separators	Components	11.0%	low (cell 0.75; not a material buyer)	12.7%	low (cell 0.97; not a material buyer)	BUILD	CHN 38%
SC-BATTERIES	Synthetic graphite (anode-grade)	Refining	6.7%	low (cell 0.94; not a material buyer)	4.6%	high 0.99	BUILD	CHN 94%
SC-BATTERIES	Li-ion cells & packs	Assembly	3.3%	low (cell 0.71; not a material buyer)	4.0%	low (cell 0.97; not a material buyer)	BUILD	CHN 84%
SC-BATTERIES	Other accumulators	Assembly	7.6%	low (cell 0.71; not a material buyer)	1.1%	low (cell 0.83; not a material buyer)	BUILD	CHN 28%
SC-EV	Electric buses, trucks & tractors	Assembly	0.0%	low 0.65	0.4%	demand 0.37	BUILD	CHN 24%
SC-EV	Electric & plug-in cars	Assembly	7.3%	low 0.65	7.0%	low 0.37	BUILD	CHN 69%
SC-EV	Electric rail vehicles	Assembly	3.1%	low n/a	10.6%	low n/a	LOW INTERVENTION	CHN 14%
SC-GRID	Copper ores & concentrates	Extraction	0.0%	high 0.98	0.0%	high 0.97	LOW INTERVENTION	CHN 13%
SC-GRID	Copper smelting & refining	Refining	6.2%	low (cell 0.71; not a material buyer)	1.7%	high 0.96	LOW INTERVENTION	CHN 50%
SC-GRID	Grain-oriented electrical steel (GOES)	Refining	25.1%	low (cell 0.71; not a material buyer)	5.2%	low (cell 0.96; not a material buyer)	BUILD	CHN 41%
SC-GRID	Insulators & bushings	Components	4.1%	low 0.60	0.8%	low (cell 0.80; not a material buyer)	BUILD	CHN 32%
SC-GRID	Power cables	Components	1.8%	low (cell 0.75; not a material buyer)	8.4%	low 0.64	LOW INTERVENTION	CHN 19%

COMPACT	NODE	STAGE	JAPAN SHARE	JAPAN NEED	KOREA SHARE	KOREA NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-GRID	Transformers	Assembly	0.0%	low (cell 0.75; not a material buyer)	9.8%	low 0.64	LOW INTERVENTION	CHN 21%
SC-GRID	Switchgear & protection	Assembly	1.8%	low (cell 0.75; not a material buyer)	3.7%	low 0.64	LOW INTERVENTION	CHN 20%
SC-HEAT	Refrigerants	Refining	1.4%	demand n/a	0.1%	demand n/a	BUILD	CHN 45%
SC-HEAT	Refrigeration & heat-pump compressors	Components	11.2%	low n/a	7.0%	low n/a	BUILD	CHN 33%
SC-HEAT	Heat pumps & AC units	Assembly	0.9%	demand n/a	1.9%	low n/a	BUILD	CHN 49%
SC-INDUSTRY	PGMs (iridium, ruthenium)	Refining	1.1%	demand n/a	0.0%	not measured n/a	LOW INTERVENTION	RUS 30%
SC-INDUSTRY	Electrolysers	Assembly	13.4%	low n/a	5.7%	low n/a	BUILD	CHN 32%
SC-INDUSTRY	Hydrogen, ammonia & methanol	Refining	0.0%	demand n/a	0.2%	demand n/a	LOW INTERVENTION	RUS 3%
SC-INDUSTRY	Furnace parts	Components	11.0%	low n/a	2.2%	low n/a	BUILD	CHN 21%
SC-INDUSTRY	Graphite electrodes	Components	7.2%	low n/a	0.0%	demand n/a	COORDINATE	CHN 38%
SC-INDUSTRY	Direct-reduced iron (DRI/HBI)	Refining	0.0%	not measured n/a	0.0%	demand n/a	LOW INTERVENTION	RUS 20%
SC-INDUSTRY	Electric furnaces & heating	Assembly	5.5%	low n/a	5.3%	low n/a	BUILD	CHN 27%
SC-MAGNETS	Rare-earth ores (monazite)	Extraction	0.0%	high 0.98	0.0%	low (cell 1.00; not a material buyer)	INSUFFICIENT DATA	CHN 59%
SC-MAGNETS	Rare-earth oxides & compounds	Refining	6.3%	high 1.00	1.0%	high 0.99	BUILD	CHN 84%
SC-MAGNETS	Rare-earth metals	Refining	2.4%	high 1.00	0.2%	low (cell 0.99; not a material buyer)	BUILD	CHN 84%
SC-MAGNETS	Rare-earth permanent magnets	Components	8.8%	demand 0.62	0.9%	high 0.97	BUILD	CHN 64%
SC-MAGNETS	Non-oriented electrical steel (NGO)	Refining	9.2%	low (cell 0.71; not a material buyer)	13.7%	low (cell 0.96; not a material buyer)	BUILD	CHN 26%
SC-MAGNETS	Ferrite & other magnets	Components	5.4%	low 0.62	4.4%	low (cell 0.97; not a material buyer)	BUILD	CHN 58%
SC-MAGNETS	Motor & generator parts	Components	7.3%	low 0.62	2.0%	low (cell 0.97; not a material buyer)	BUILD	CHN 26%
SC-MAGNETS	Electric motors	Assembly	5.5%	low 0.62	2.6%	high 0.97	BUILD	CHN 26%
SC-MAGNETS	AC generators	Assembly	1.5%	low 0.60	0.3%	high 0.80	LOW INTERVENTION	CHN 15%
SC-NUCLEAR	Uranium conversion & enrichment	Refining	0.1%	low (cell 0.97; not a material buyer)	0.0%	high 0.98	LOW INTERVENTION	RUS 16%
SC-NUCLEAR	Fuel assemblies & zirconium	Components	0.7%	demand 0.61	7.4%	low 0.54	LOW INTERVENTION	RUS 28%
SC-NUCLEAR	Nuclear special materials	Refining	4.8%	low n/a	0.1%	demand n/a	LOW INTERVENTION	0%
SC-SEMIS	Piezo-electric quartz	Refining	10.8%	demand n/a	0.0%	not measured n/a	BUILD	CHN 60%

COMPACT	NODE	STAGE	JAPAN SHARE	JAPAN NEED	KOREA SHARE	KOREA NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-SEMIS	Gallium, germanium & indium	Refining	3.3%	demand n/a	18.0%	low n/a	BUILD	CHN 25%
SC-SEMIS	Semiconductor manufacturing equipment	Assembly	21.2%	low 0.55	5.6%	demand 0.25	LOW INTERVENTION	CHN 5%
SC-SEMIS	Integrated circuits	Components	5.2%	low n/a	17.1%	low n/a	LOW INTERVENTION	CHN 16%
SC-SEMIS	Power semiconductors	Components	6.2%	low 0.35	3.2%	low (cell 0.70; not a material buyer)	BUILD	CHN 32%
SC-SEMIS	Sensors & controller parts	Components	8.6%	demand n/a	2.4%	low n/a	BUILD	CHN 24%
SC-SEMIS	Power electronics components	Components	8.6%	low 0.62	2.5%	low (cell 0.97; not a material buyer)	BUILD	CHN 37%
SC-SEMIS	Automatic controllers	Assembly	11.3%	low (cell 0.67; not a material buyer)	2.8%	demand 0.51	LOW INTERVENTION	CHN 12%
SC-SEMIS	Static converters & inverters	Assembly	4.2%	low (cell 0.67; not a material buyer)	1.3%	low 0.51	BUILD	CHN 38%
SC-SOLAR	Polysilicon	Refining	7.0%	high 0.95	1.2%	high 0.71	LOW INTERVENTION	CHN 97%
SC-SOLAR	Wafers	Refining	28.4%	low 0.32	6.4%	demand 0.30	BUILD	CHN 98%
SC-SOLAR	Silver powder	Refining	59.1%	low n/a	1.6%	demand n/a	LOW INTERVENTION	0%
SC-SOLAR	Solar cells	Components	0.3%	low 0.32	11.1%	low 0.30	BUILD	CHN 92%
SC-SOLAR	PV cover glass	Components	0.2%	low 0.32	1.9%	demand 0.30	BUILD	CHN 61%
SC-SOLAR	PV modules & generators	Assembly	0.1%	demand 0.55	0.0%	low 0.25	BUILD	CHN 84%
SC-WIND	Wind blades & hubs	Components	2.8%	low 0.60	1.1%	low (cell 0.80; not a material buyer)	BUILD	CHN 71%
SC-WIND	Wind towers	Components	0.5%	low 0.60	11.7%	low (cell 0.80; not a material buyer)	LOW INTERVENTION	CHN 18%
SC-WIND	Hydro turbine parts	Components	0.6%	demand n/a	0.0%	not measured n/a	COORDINATE	CHN 35%
SC-WIND	Wind turbine generating sets	Assembly	0.0%	high 0.75	0.1%	low 0.64	BUILD	CHN 74%

Sources: 1.

Table A2. Section 3, Bilateral Complementarity Map

RANK	TIER	TECHNOLOGY	UNIT	STAGE	JAPAN POSITION	KOREA POSITION	DYADIC RELATIONSHIP	STRATEGIC CASE
1	joint bet	Solar Manufacturing	Wafers	Refining	strong (28.4% of world supply)	moderate (6.4% of world supply)	Capacity-constrained trusted supply	geopolitical chokepoint; need 0.32 / 0.30
2	joint bet	Batteries	Batteries refining (battery materials (foil, electrolyte), cathode materials & precursors, cobalt intermediates & chemicals)	Refining	moderate	weak	Shared vulnerability	geopolitical chokepoint; need 0.94 / 0.99
3	joint bet	Electric Vehicles	Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds)	Refining	moderate	weak	Capacity-constrained trusted supply	geopolitical chokepoint; need 1.00 / 0.99
4	joint bet	Wind	Wind refining (copper wire & rod, rare-earth metals, rare-earth oxides & compounds)	Refining	weak	weak	Shared vulnerability	geopolitical chokepoint; need 0.69 / 0.75
5	joint bet	Permanent Magnets & Drive Systems	Rare-earth oxides & compounds	Refining	moderate (6.3% of world supply)	weak (1.0% of world supply)	Shared vulnerability	geopolitical chokepoint; need 1.00 / 0.99
6	joint bet	Batteries	Batteries extraction (cobalt ores, lithium minerals, natural graphite)	Extraction	weak	weak	Shared vulnerability	geopolitical chokepoint; need 0.98 / 1.00
7	joint bet	Battery Materials & Cells	Natural graphite	Extraction	moderate (1.7% of world supply)	moderate (2.6% of world supply)	Shared vulnerability	geopolitical chokepoint; need 0.98 / 1.00

RANK	TIER	TECHNOLOGY	UNIT	STAGE	JAPAN POSITION	KOREA POSITION	DYADIC RELATIONSHIP	STRATEGIC CASE
8	joint bet	Electric Vehicles	Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite)	Extraction	weak	weak	Shared vulnerability	geopolitical chokepoint; need 0.98 / 1.00
9	joint bet	Permanent Magnets & Drive Systems	Rare-earth permanent magnets	Components	strong (8.8% of world supply)	weak (0.9% of world supply)	Capacity-constrained trusted supply	geopolitical chokepoint; need 0.62 / 0.97
10	joint bet	Battery Materials & Cells	Cobalt intermediates & chemicals	Refining	moderate (2.4% of world supply)	moderate (0.8% of world supply)	Shared vulnerability	geopolitical chokepoint; need 0.94 / 0.99
11	joint bet	Battery Materials & Cells	Lithium chemicals	Refining	moderate (0.1% of world supply)	moderate (0.7% of world supply)	Shared vulnerability	geopolitical chokepoint; need 0.94 / 0.99
12	joint bet	Electric Grid	Electric Grid refining (copper smelting & refining, copper wire & rod, grain-oriented electrical steel (goes))	Refining	moderate	weak	Shared vulnerability	geopolitical chokepoint; need 0.71 / 0.96
13	joint bet	Battery Materials & Cells	Battery materials (foil, electrolyte)	Refining	strong (11.2% of world supply)	moderate (9.5% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need 0.94 / 0.99
14	joint bet	Coal	Coal refining	Refining	moderate	weak	Shared vulnerability	strategic manufacturing gap; need 0.83 / 0.93
15	joint bet	Batteries	Batteries components (accumulator parts & separators)	Components	moderate	moderate	Shared vulnerability	strategic manufacturing gap; need 0.75 / 0.73
16	joint bet	Electric Grid	Electric Grid components (insulators & bushings, power electronics components, power semiconductors)	Components	strong	moderate	Japan supplies / Korea needs	strategic manufacturing gap; need 0.35 / 0.70
17	coordinate	Solar Manufacturing	Polysilicon	Refining	moderate (7.0% of world supply)	moderate (1.2% of world supply)	Shared vulnerability	geopolitical chokepoint; need 0.95 / 0.71
18	coordinate	Solar Manufacturing	Solar cells	Components	moderate (0.3% of world supply)	strong (11.1% of world supply)	Complementary specialisation	geopolitical chokepoint; need 0.32 / 0.30
19	coordinate	Battery Materials & Cells	Cathode materials & precursors	Refining	moderate (9.7% of world supply)	strong (50.9% of world supply)	Complementary specialisation	geopolitical chokepoint; need 0.94 / 0.99
20	coordinate	Nuclear	Nuclear components (fuel assemblies & zirconium, reactor parts)	Components	moderate	strong	Complementary specialisation	strategic manufacturing gap; need 0.61 / 0.54
21	coordinate	Solar	Solar refining (copper wire & rod, polysilicon)	Refining	moderate	weak	Shared vulnerability	geopolitical chokepoint; need 0.95 / 0.71
22	coordinate	Semiconductors & Power Electronics	Piezo-electric quartz	Refining	strong (10.8% of world supply)	weak (0.0% of world supply)	Complementary specialisation	geopolitical chokepoint; need n/a / n/a
23	coordinate	Gas	Gas components	Components	strong	moderate	Complementary specialisation	strategic manufacturing gap; need 0.59 / 0.50
24	coordinate	Batteries	Batteries assembly (li-ion cells & packs, other accumulators)	Assembly	moderate	strong	Korea supplies / Japan needs	geopolitical chokepoint; need 0.71 / 0.83
25	coordinate	Battery Materials & Cells	Manganese dioxide (EMD)	Refining	strong (18.3% of world supply)	weak (0.0% of world supply)	Complementary specialisation	trusted partner capacity constraint; need 0.94 / 0.99
26	coordinate	Semiconductors & Power Electronics	Gallium, germanium & indium	Refining	moderate (3.3% of world supply)	strong (18.0% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need n/a / n/a
27	coordinate	Grid Equipment & Conductors	Grain-oriented electrical steel (GOES)	Refining	strong (25.1% of world supply)	moderate (5.2% of world supply)	Complementary specialisation	trusted partner capacity constraint; need 0.71 / 0.96
28	coordinate	Permanent Magnets & Drive Systems	Non-oriented electrical steel (NGO)	Refining	moderate (9.2% of world supply)	strong (13.7% of world supply)	Complementary specialisation	trusted partner capacity constraint; need 0.71 / 0.96
29	coordinate	Semiconductors & Power Electronics	Automatic controllers	Assembly	strong (11.3% of world supply)	moderate (2.8% of world supply)	Complementary specialisation	strategic manufacturing gap; need 0.67 / 0.51
30	coordinate	Semiconductors & Power Electronics	Integrated circuits	Components	moderate (5.2% of world supply)	strong (17.1% of world supply)	Complementary specialisation	strategic manufacturing gap; need n/a / n/a

RANK	TIER	TECHNOLOGY	UNIT	STAGE	JAPAN POSITION	KOREA POSITION	DYADIC RELATIONSHIP	STRATEGIC CASE
31	coordinate	Semiconductors & Power Electronics	Semiconductor manufacturing equipment	Assembly	strong (21.2% of world supply)	moderate (5.6% of world supply)	Complementary specialisation	strategic manufacturing gap; need 0.55 / 0.25
32	coordinate	Solar Manufacturing	Silver powder	Refining	strong (59.1% of world supply)	weak (1.6% of world supply)	Complementary specialisation	strategic manufacturing gap; need n/a / n/a
33	coordinate	Grid Equipment & Conductors	Power cables	Components	weak (1.8% of world supply)	strong (8.4% of world supply)	Complementary specialisation	strategic manufacturing gap; need 0.75 / 0.64
34	manage competition	Nuclear	Nuclear assembly (nuclear reactors & enrichment machinery)	Assembly	strong	strong	Direct competition	no material intervention; need 0.66 / 0.62
35	manage competition	Gas	Gas assembly	Assembly	strong	strong	Direct competition	no material intervention; need 0.40 / 0.34
36	manage competition	Battery Materials & Cells	Accumulator parts & separators	Components	strong (11.0% of world supply)	strong (12.7% of world supply)	Direct competition	trusted partner capacity constraint; need 0.75 / 0.97
37	manage competition	Coal	Coal components	Components	strong	strong	Direct competition	no material intervention; need 0.71 / 0.58
38	manage competition	Oil	Oil assembly (ac generators)	Assembly	strong	strong	Direct competition	no material intervention; need 0.36 / 0.30
39	manage competition	Coal	Coal assembly	Assembly	strong	strong	Direct competition	no material intervention; need 0.59 / 0.60

Sources: 1.

Table A3. Section 4, Trade Agenda

ID	DIRECTION	UNIT	ACTION TYPE	ACTION
T01	JPN -> KOR	wafers	tariff risk	Bind market access for Japan Wafers into Korea: Korea's protection priority on this node is in its top third (0.81), which taxes the trusted supplier it needs.
T02	both	Batteries/Refining	anti circumvention	Common origin and anti-circumvention rules for Batteries refining (battery materials (foil, electrolyte), cathode materials & precursors, cobalt intermediates & chemicals): both import heavily from CHN, so a measure one side takes alone leaks through the other.
T03	JPN -> KOR	Electric Vehicles/Refining	procurement recognition	Recognise Japan output of Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds) as trusted domestic-equivalent supply in Korea procurement and content rules; capacity, not access, is the constraint.
T04	both	Wind/Refining	anti circumvention	Common origin and anti-circumvention rules for Wind refining (copper wire & rod, rare-earth metals, rare-earth oxides & compounds): both import heavily from CHN, so a measure one side takes alone leaks through the other.
T05	both	rare-earth-oxides-and-compounds	anti circumvention	Common origin and anti-circumvention rules for Rare-earth oxides & compounds: both import heavily from CHN, so a measure one side takes alone leaks through the other.
T06	both	Batteries/Extraction	common standards	Common specifications for Batteries extraction (cobalt ores, lithium minerals, natural graphite) so a joint supplier base can serve both markets.
T07	both	natural-graphite	common standards	Common specifications for Natural graphite so a joint supplier base can serve both markets.
T08	both	Electric Vehicles/Extraction	common standards	Common specifications for Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite) so a joint supplier base can serve both markets.
T09	JPN -> KOR	rare-earth-permanent-magnets	tariff risk	Bind market access for Japan Rare-earth permanent magnets into Korea: Korea's protection priority on this node is in its top third (0.86), which taxes the trusted supplier it needs.
T10	both	cobalt-intermediates-and-chemicals	anti circumvention	Common origin and anti-circumvention rules for Cobalt intermediates & chemicals: both import heavily from CHN, so a measure one side takes alone leaks through the other.
T11	both	lithium-chemicals	anti circumvention	Common origin and anti-circumvention rules for Lithium chemicals: both import heavily from CHN, so a measure one side takes alone leaks through the other.
T12	both	Electric Grid/Refining	common standards	Common specifications for Electric Grid refining (copper smelting & refining, copper wire & rod, grain-oriented electrical steel (goes)) so a joint supplier base can serve both markets.
T13	open	pgms-iridium-ruthenium	open sourcing	Leave PGMs (iridium, ruthenium) open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.

ID	DIRECTION	UNIT	ACTION TYPE	ACTION
T14	open	copper-ores-and-concentrates	open sourcing	Leave Copper ores & concentrates open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T15	open	nickel-intermediates-and-chemicals	open sourcing	Leave Nickel intermediates & chemicals open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T16	open	nuclear-special-materials	open sourcing	Leave Nuclear special materials open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T17	open	battery-recycling-feed	open sourcing	Leave Battery recycling feed open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T18	open	lithium-minerals	open sourcing	Leave Lithium minerals open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T19	both	Nuclear/Assembly	no preference	No bilateral preference for Nuclear assembly (nuclear reactors & enrichment machinery): both are strong suppliers; keep the market contestable.
T20	both	Gas/Assembly	no preference	No bilateral preference for Gas assembly: both are strong suppliers; keep the market contestable.
T21	both	accumulator-parts-and-separators	no preference	No bilateral preference for Accumulator parts & separators: both are strong suppliers; keep the market contestable.
T22	both	Coal/Components	no preference	No bilateral preference for Coal components: both are strong suppliers; keep the market contestable.
T23	both	Oil/Assembly	no preference	No bilateral preference for Oil assembly (ac generators): both are strong suppliers; keep the market contestable.
T24	both	Coal/Assembly	no preference	No bilateral preference for Coal assembly: both are strong suppliers; keep the market contestable.

Sources: 1.

Rule-generated from the data (R/partnerships) with external facts from a dated, curated evidence layer; not yet checked by an independent verifier. Items marked partial or unverified must be confirmed before citation.

Allied Industrial Base Explorer, Partnership memo: Japan–Korea Strategic Energy & Industrial Partnership